

Sheetname: CPU

CPU MC68000  
CLOCK 16MHZ  
DATA AND ADDRESS BUFFERS  
HALT AND RESET  
DTACK GLUE LOGIC  
INTERRUPT

Sheetname: MEMORY

ROM 16kBytes  
RAM 1MBytes

Sheetname: FOUR UARTS

TG16C554  
4 UARTS.

Arquivo: fourUarts.kicad\_sch

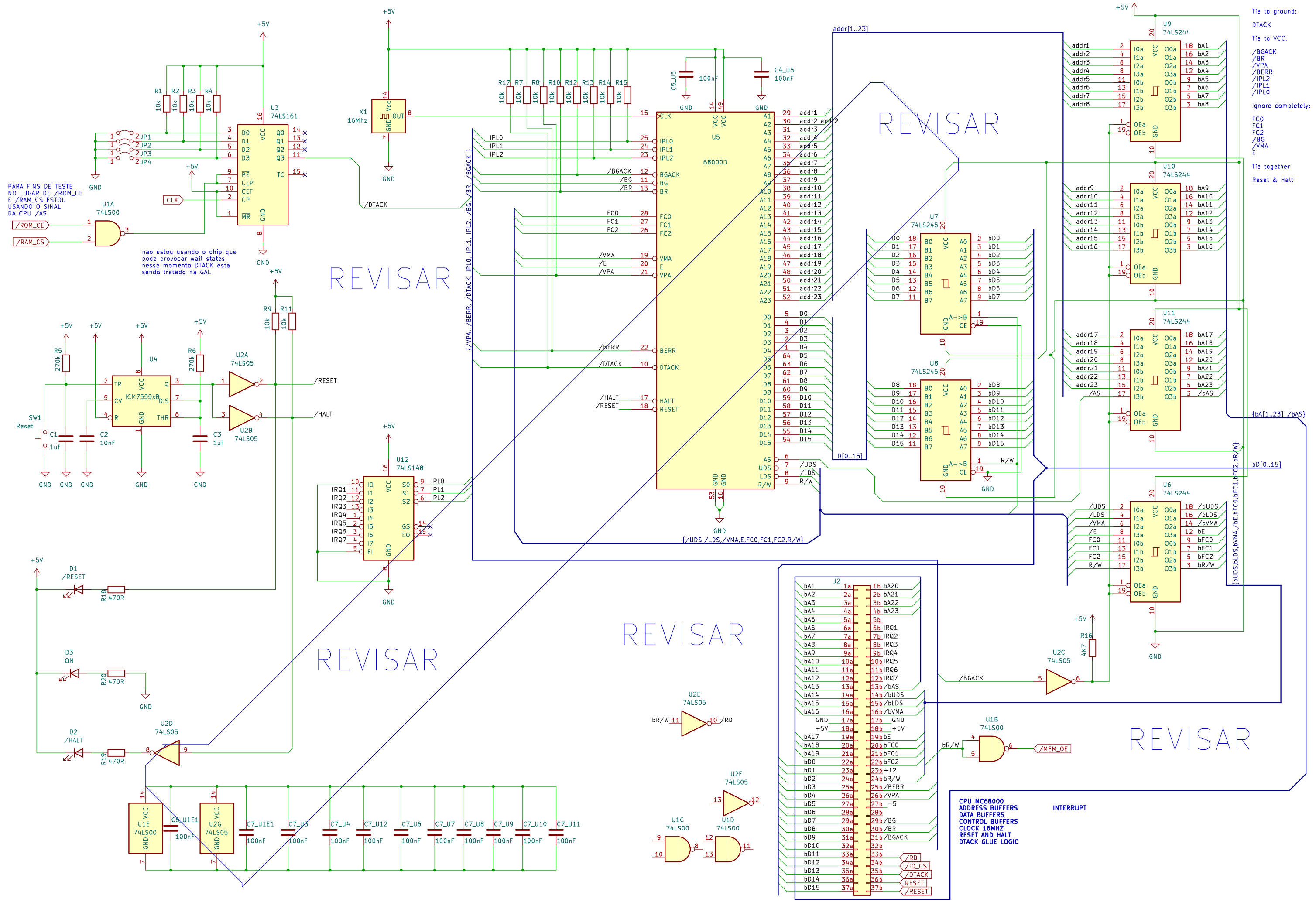
Sheetname: cpuNoBuffer

CPU MC68000  
CLOCK 16MHZ  
HALT AND RESET  
DTACK GLUE LOGIC  
INTERRUPT

Arquivo: cpuNoBuffer.kicad\_sch

UART CONTROL

Arquivo: uart\_control.kicad\_sch



PARA FINS DE TESTE  
NO LUGAR DE /ROM\_CE  
E /RAM\_CS ESTOU  
USANDO O SINAL  
DA CPU /AS

nao estou usando o chip que  
pode provocar wait states  
nesse momento DTACK está  
sendo tratado na GAL

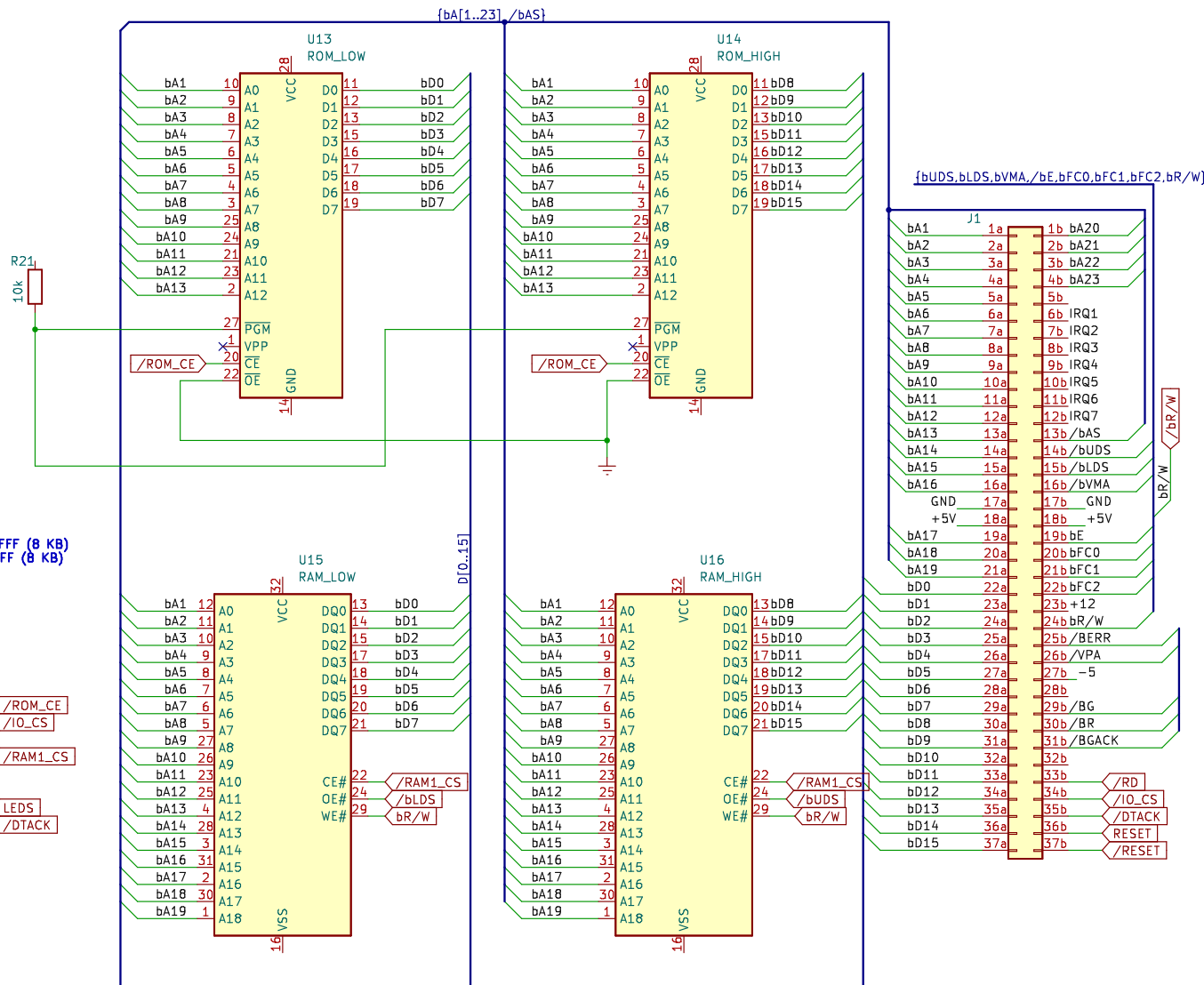
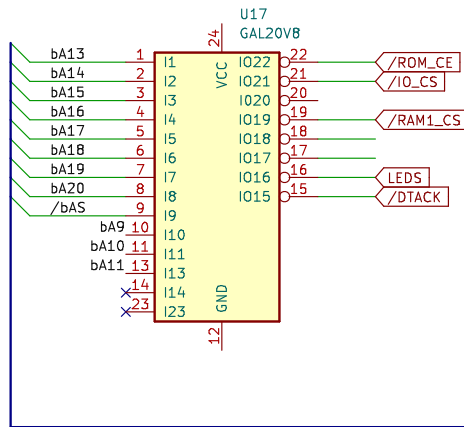
CPU MC68000  
ADDRESS BUFFERS  
DATA BUFFERS  
CONTROL BUFFERS  
CLOCK 16MHZ  
RESET AND HALT  
DTACK GLUE LOGIC

Tie to ground:  
DTACK  
Tie to VCC:  
/BGACK  
/BR  
/VPA  
/BERR  
/IPL2  
/IPL1  
/IPL0  
Ignore completely:  
FC0  
FC1  
FC2  
/BG  
/VMA  
E  
Tie together  
Reset & Halt

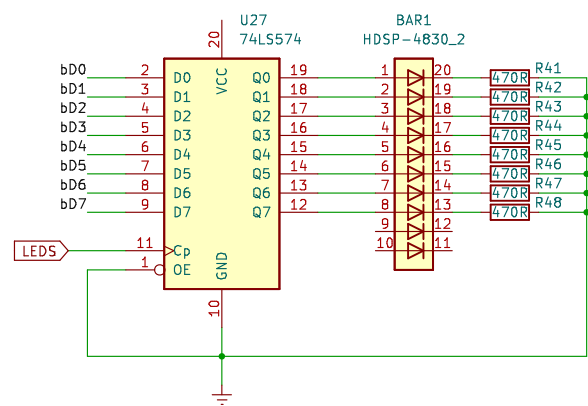
; PINOS DE ENTRADA  
A13 PIN 1 : Barramento de endereço  
A14 PIN 2  
A15 PIN 3  
A16 PIN 4  
A17 PIN 5  
A18 PIN 6  
A19 PIN 7  
A20 PIN 8  
IAS PIN 9 : Address Strobe

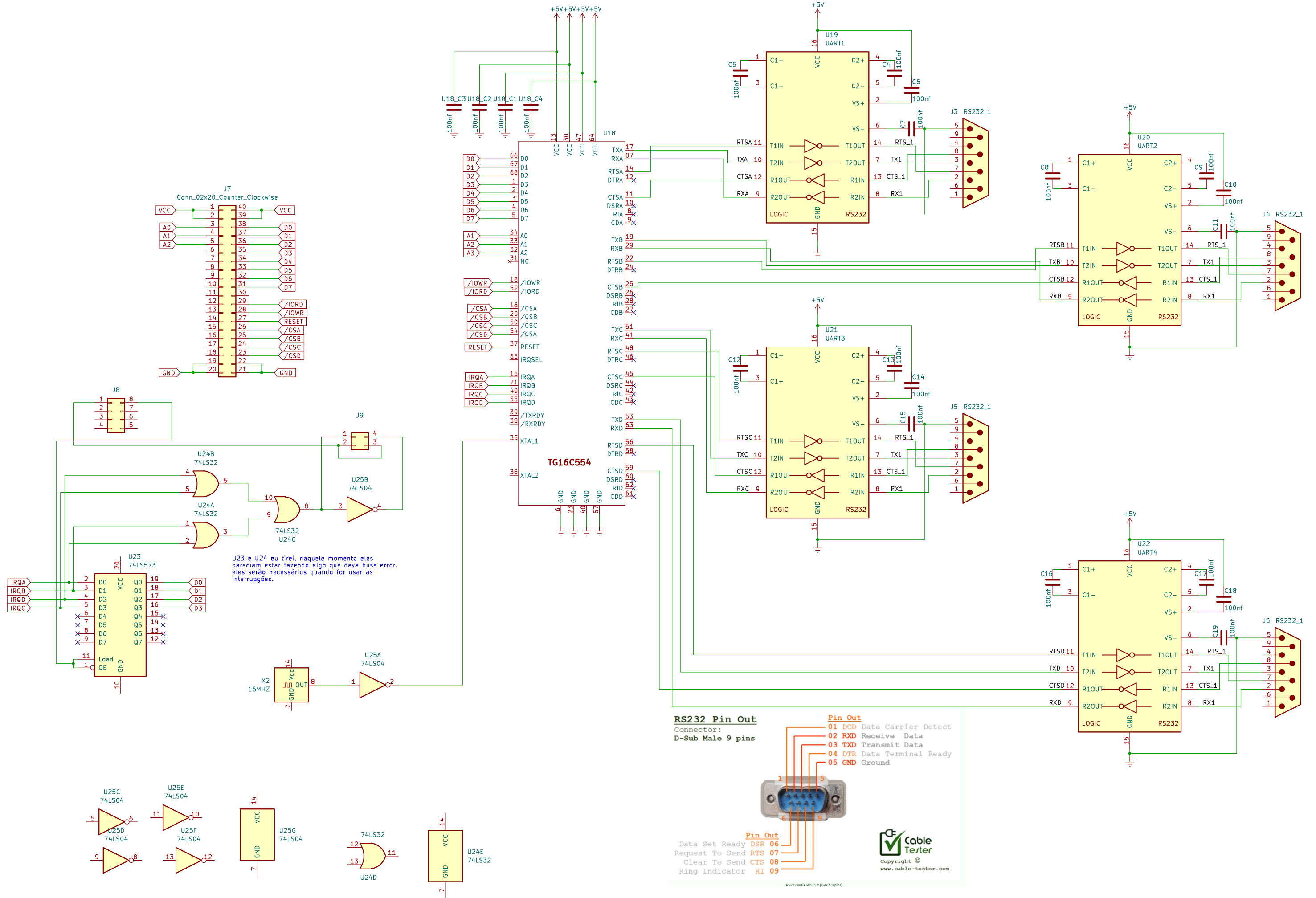
; PINOS DE SAÍDA  
IROM\_CS PIN 19 : ROM (0x00000-0x01FFF)  
IIQ\_CS PIN 18 : I/O (0x02000-0x03FFF)  
IRAM4\_CS PIN 17 : RAM1 (0x004000-0x07FFFF)  
IRAM1\_CS PIN 16 : RAM2 (0x080000-0x0FFFFFF)  
IRAM2\_CS PIN 16 : RAM2 (0x100000-0x17FFFF)  
IRAM3\_CS PIN 16 : RAM2 (0x180000-0x1FFFFFF)

; EQUACÕES  
IROM\_CS = IAS & IA20 & IA19 & IA18 & IA17 & IA16 & IA15 & IA14 & IA13 : 0x0000-0x1FFF (8 KB)  
IIQ\_CS = IAS & IA20 & IA19 & IA18 & IA17 & IA16 & IA15 & IA14 & A13 : 0x2000-0x3FFF (8 KB)  
IRAM4\_CS = IAS & IA20 & IA19 : 0x004000-0x07FFFF (20=0 e A19=0, exceto ROM e I/O)  
IRAM1\_CS = IAS & IA20 & A19 : 0x080000-0x0FFFFFF (20=0 e A19=1)  
IRAM2\_CS = IAS & A20 & A19 : 0x100000-0x17FFFF (20=1 e A19=0)  
IRAM3\_CS = IAS & A20 & A19 : 0x180000-0x1FFFFFF (20=1 e A19=1)

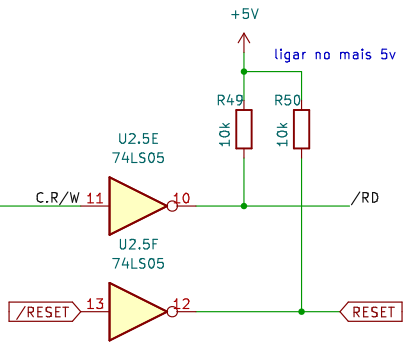
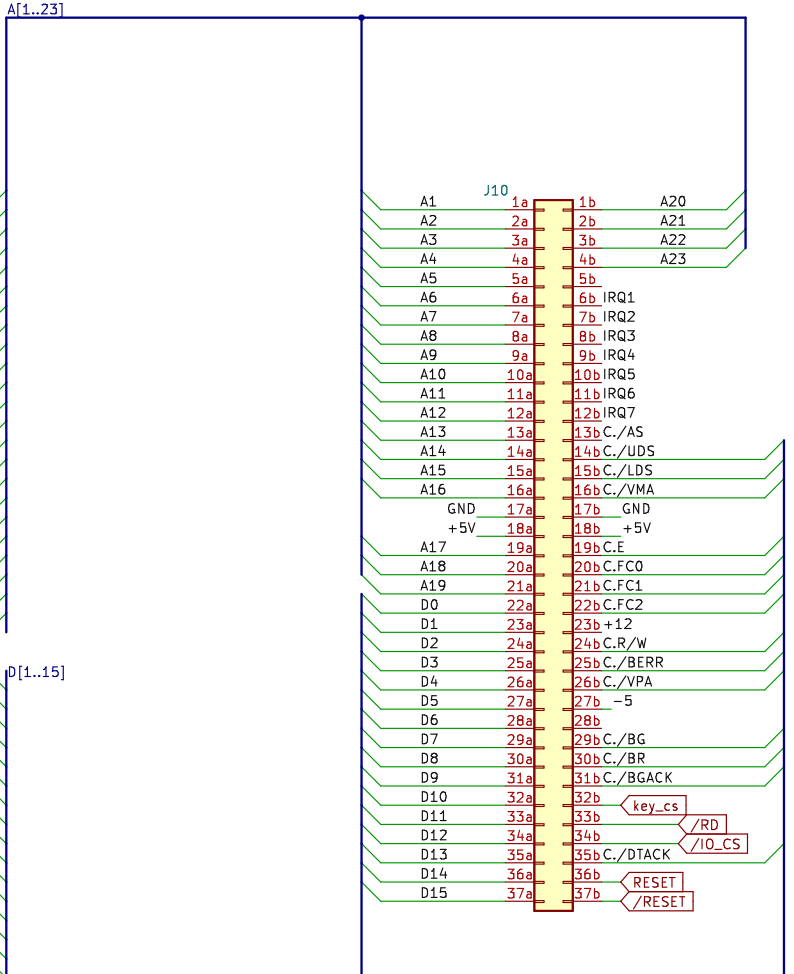
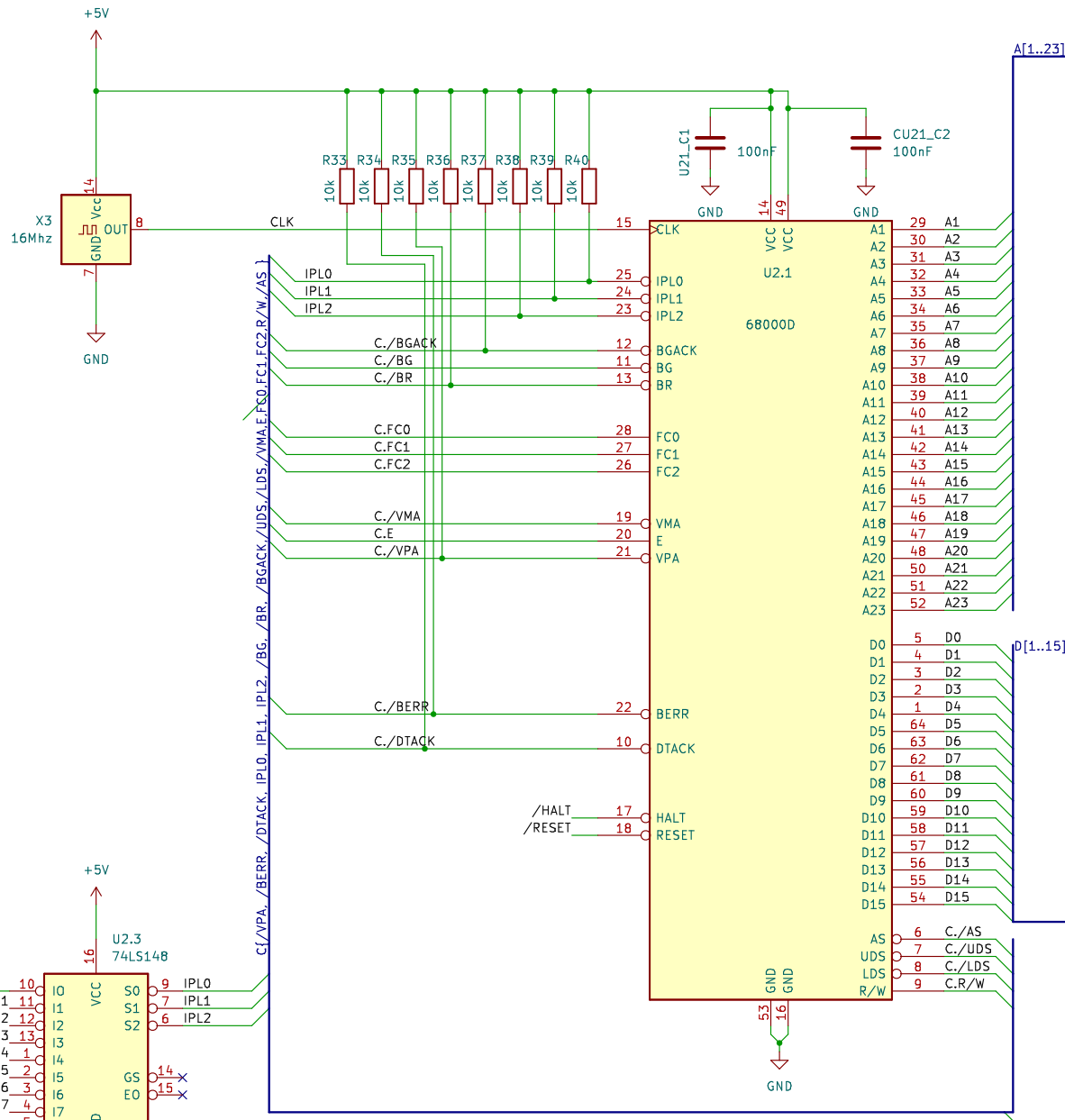
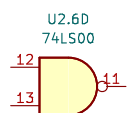
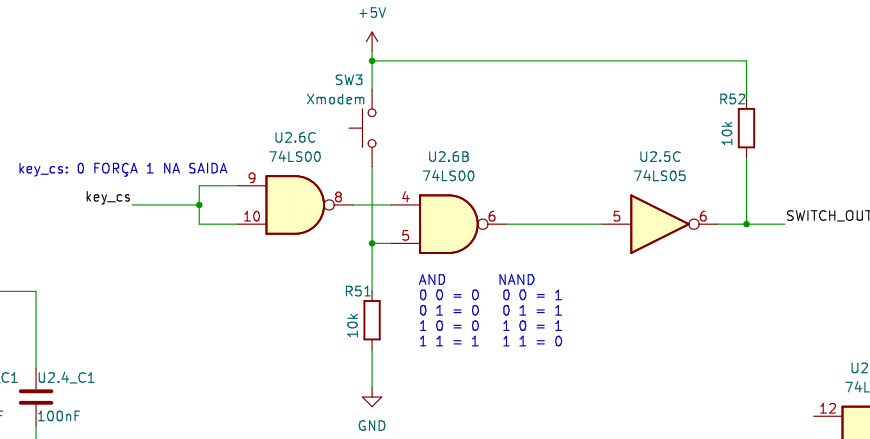
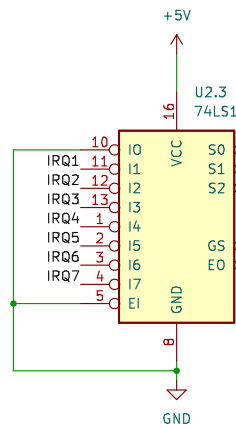
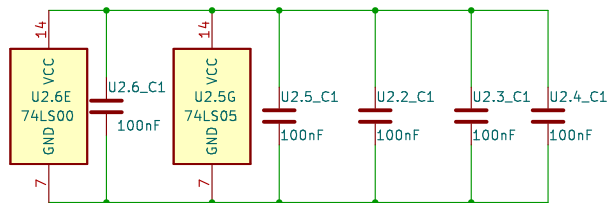
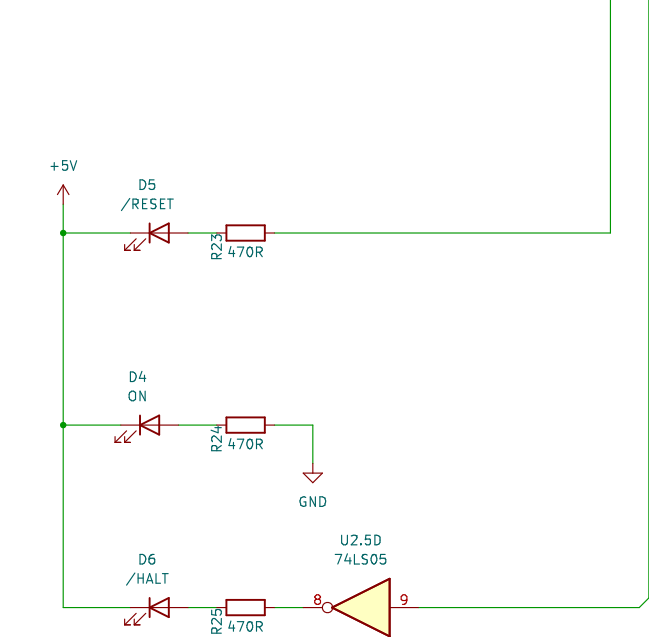
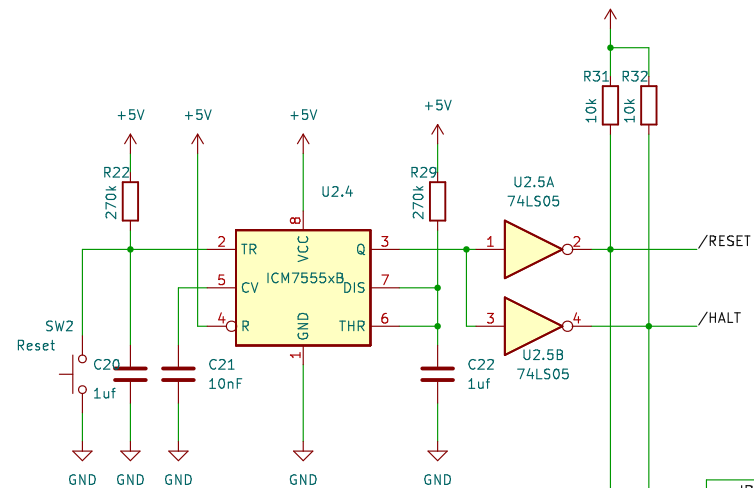
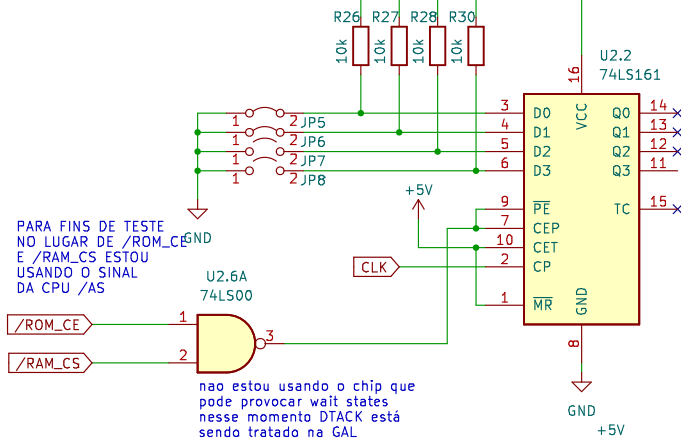


**MEMORY CARD**  
**16KB ROM**  
**1MB RAM**





- 0 - por o soquete do ci intrq
- 1 - por todos os capacitores 100nf
- 2 - por todos os terras
- 3 - por todos os vccs
- 4 - por todos os pullups com opção de pulldown
- 5 - por o oscilador
- 6 - por o ci intrq
- 7 - ligar o clock
- 8 - ligar a cpu ao halt reset
- 9 - ligar os leds



CPU MC68000  
INTERRUPT  
CLOCK 16MHZ  
RESET AND HALT  
DTACK GLUE LOGIC

UART1 - 0x2000 até 0x20FF = UART\_CS1  
UART2 - 0x2100 até 0x21FF = UART\_CS2  
UART3 - 0x2200 até 0x22FF = UART\_CS3  
UART4 - 0x2300 até 0x23FF = UART\_CS4

